SUKHMANPREET **KAUR**

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Summary

Biology researcher at CSUSB studying influenza, contributing to microalgal biomass research at UCR, and joining the CIRM research program in 2027, with a goal of becoming a biology professor.

Education and Training

Master in biology — **Present**
California State University, San Bernardino *(Pursuing MS Biology)* — San Bernardino, CA
Bachelor of Science: **Biology** — **05/2023**
Guru Nanak Dev University *(Graduated with Distinction)* — India

Skills

Molecular & Cellular Biology Techniques: Cell culture and maintenance, cell counting (manual & automated), cryopreservation, freezing/thawing cells, RNA extraction and purification, qRT-PCR, gel electrophoresis, cDNA synthesis / reverse transcription plasmid construction, BSL-2 practices (PPE, decontamination, virus handling)
Microscopy & Instrumentation: Light and fluorescence microscopy, inverted and compound microscopes, NanoDrop 2000, Bio-Rad TC20 Automated Cell Counter
Field & Ecological Skills: Plant species identification, plant conservation data collection, GPS/GIS for land and vegetation monitoring
Research & Professional Skills: Scientific communication, collaboration, problem-solving, accurate data recording, attention to detail
Software & Data Tools: NanoDrop software, Image Lab, Microsoft Office Suite, Google Workspace, basic data analysis

Experience

Independent Researcher – Molecular Virology at CSUSB — **09/2025 to Current**
San Bernardino, CA
- Investigating NP-peptide inhibition of influenza A viral mRNA in A549 cells
- Performing BSL-2 viral infection assays using MOI-controlled inocula and stable GFP/NP-peptide cell lines
- Extracting and evaluating RNA quality with Trizol, NanoDrop metrics, and bleach agarose gels
- Conducting cDNA synthesis and SYBR-based qRT-PCR to quantify viral gene expressionPerforming BSL-2 viral infection assays using MOI-controlled inocula and stable GFP/NP-peptide cell lines

Undergraduate Research – Department of Ecology, and Organismal Biology at UCR — **06/2025 to Current**
Riverside, CA
- Processed ~155 river water samples to assess microalgal biomass in the Santa Ana River
- Identified and counted ~23,000 algal cells using 40× microscopy and hemocytometer analysis
- Measured algal cell size and maintained organized sample records for biomass analysis

Minerva Hoyt Internship – Joshua Tree National Park — **01/2024 to 05/2025**
Joshua Tree, CA
- Performed data collection on various plant and animal species within the confines of JNTP
- Enhanced accuracy in data entry, ensuring well-organized records
- Collected necessary details for research endeavors and documentation
- Engaged in educational programs to enhance conservation and environmental awareness
- Collaborated with Joshua Tree National Park employees in Science and Resource Stewardship Division

Science Tutor (**Self-employed)** — **06/2023 to 07/2024**
Gurdaspur, Punjab
- Instructed a diverse group of 12 high school students in Chemistry and Biology
- Assessed student mastery through tests, quizzes, and hands-on experiments
- Clarified complex science with visuals, analogies, and real-world examples

Volunteer

Volunteer @ Mojave Desert Land Trust (MDLT) - Joshua Tree, CA
- Supported habitat restoration, invasive species removal, field surveys, data collection, and community outreach to advance land conservation and environmental awareness at Mojave Desert Land Trust (MDLT).

Accomplishments

- Sally Casanova Pre-Doctoral Scholar: Awarded the Sally Casanova Pre-Doctoral Scholarship for the 2026–2027
- CIRM Internship Program : Candidate for the Summer 2027 Cohort.
- Poster presented: NP peptide Inhibition of Influenza Virus mRNA, CSUSB Research Symposium, 2025
- Minerva Hoyt Intern 2025: Selected in a competitive internship program focused on plant conservation and research
- Featured artist in the 2025 edition of HOWL Magazine. Awarded for excellence in visual art by Copper Mountain College
- University Youth Festival 2022 and 2023: Awarded third place for outstanding performance in painting flora
- Mohindra Pride 2022: Awarded for outstanding public speaking
- Biological Society Flower Show 2023: Recognized for innovative representation of fossilized plants

Personal Interests: Hiking • Reading and Research • Volunteering • Nature Journaling • painting• Singing